A φ-Ladder Prediction for Neutrino Mass Ordering and the Dimensionless Splitting Ratio

A Falsifiable Prediction from the Recognition Science Mass Backbone

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Abstract

The Recognition Science mass backbone places neutrinos on deep negative rungs of the φ-ladder ($\varphi = (1+\sqrt{5})/2$), with fractional rung positions forced by octave compatibility and the edge-partition structure of the three-cube Q_3. Given the neutral-sector quarter-step structure, the framework makes three predictions that are independent of the SI calibration seam:

1. **Mass ordering**: normal ($m_1 < m_2 < m_3$), forced by the monotonicity of the φ-ladder.
2. **Squared-mass ratio**: $m_3^2/m_2^2 = \varphi^7 \approx 29.03$.
3. **Splitting ratio**: $R_\Delta := \Delta m_{31}^2/\Delta m_{21}^2 = (\varphi^{11}-1)/(\varphi^4-1) \approx 33.82$.

The splitting ratio is the sharpest near-term test. It is dimensionless, depends only on the golden ratio and derived cube integers ($E_{\text{pass}} = 11$, $2^{D-1} = 4$), and carries no dependence on the absolute mass scale or the SI anchor. The current NuFIT 5.3 best fit for normal ordering gives $R_\Delta^{\text{exp}} = 31.6 \pm 1.5$, placing the prediction at approximately 1.5σ tension. If future oscillation data push the measured R_Δ beyond 3σ from 33.82, the framework's neutrino rung assignment is refuted.

One structural input remains at the frontier: the $-1/4$ quarter-phase offset that selects the neutrino rung triple from the deep-ladder window. This paper frames the prediction conditionally on that offset rather than claiming it as a closed first-principles derivation.

1 Introduction

The neutrino mass ordering—whether the lightest neutrino mass eigenstate is ν_1 (normal ordering) or ν_3 (inverted ordering)—is one of the major open questions in particle physics. Current oscillation experiments prefer normal ordering but have not established it decisively. The absolute neutrino mass scale remains unknown, bounded from above by cosmological constraints ($\Sigma m_\nu < 0.12$ eV) and tritium endpoint measurements.

A framework that predicts the mass ordering, the splitting ratio, and the absolute mass scale from a discrete combinatorial structure would be either a powerful tool or a conspicuous target for falsification. This paper provides such predictions from the Recognition Science mass backbone.

The backbone paper (Paper I [**?**]) established the fermion mass law at the common anchor scale $\mu_\star = 182.201$ GeV, with sector yardsticks, integer rungs, and the charge-band correction all derived from the three-cube Q_3. The lepton paper (Paper II [**?**]) demonstrated charged-lepton mass ratios from Q_3 integers with no free parameters. The CKM paper (Paper III [**?**]) showed how the backbone connects to Standard Model mixing observables.

The present paper applies the backbone to the neutrino sector. The key simplification is that neutrinos are electrically neutral ($Q = 0$), so the charge-band correction vanishes identically: $\mathrm{gap}(0) = 0$. The neutrino mass spectrum reduces to a pure φ-ladder, and every mass ratio is a φ-power of a rung difference.

This simplification makes the neutrino sector the cleanest testing ground for the framework. The dimensionless splitting ratio R_Δ depends on nothing but φ and the rung differences, which are derived from cube integers. It is independent of the absolute mass scale, the sector yardstick, the SI calibration seam, and even the electroweak VEV.

1.1 Scope and framing

This paper is a *prediction paper*, not a full-derivation paper. It states the neutrino predictions that follow from the neutral-sector quarter-step structure and identifies the one structural input that remains at the frontier. The predictions are conditional on that input; they are sharp, falsifiable, and honest about what is assumed versus derived.

What belongs in this paper: the neutral-sector simplification, the quarter-step lattice, the edge-partition rung spacings, the absolute baseline derivation, and the three seam-free predictions (ordering, φ^7 ratio, splitting ratio).

What does not belong: the full RS ontology, the charged-lepton derivation, quark masses, or the anchor/units argument beyond a brief citation of Paper I.

2 Notation and prerequisites

We use the backbone conventions of Paper I [**?**]. The cube integers are $V = 8$, $E = E_{\mathrm{total}} = 12$, $F = 6$, $A = 1$, $E_{\mathrm{pass}} = E - A = 11$, $W = E_{\mathrm{pass}} + F = 17$. The golden ratio is $\varphi = (1+\sqrt{5})/2$, forced by the Recognition Composition Law. The octave period is $T_{\min} = 2^D = 8$ for spatial dimension $D = 3$.

The mass law at the anchor scale $\mu_\star$ (Paper I, Eq. (1)) is

$$m_f^{\mathrm{RS}}(\mu_\star) = A_{s(f)}\, \varphi^{\,r_f - 8 + \mathrm{gap}(Z_f)}, \tag{1}$$

where A_s is the sector yardstick and $\mathrm{gap}(Z) = \log_\varphi(1 + Z/\varphi)$ is the charge-band correction.

3 Neutral-sector simplification

Proposition 1 (Neutral-sector mass law)**.** *For neutrinos ($Q_\nu = 0$), the mass law reduces to*

$$m_{\nu_i} = A_{\mathrm{EW}}\, \varphi^{\,r_i - 8}, \tag{2}$$

with no charge-band correction.

Proof. $Q_\nu = 0$ implies $\tilde{Q} = 6Q = 0$, hence $Z_\nu = \tilde{Q}^2 + \tilde{Q}^4 = 0$, and therefore $\mathrm{gap}(0) = \log_\varphi(1 + 0/\varphi) = \log_\varphi 1 = 0$. The neutrino sector yardstick is A_{EW} (electroweak sector: $B_{\mathrm{pow}} = +1$, $r_0 = 55$), giving $A_{\mathrm{EW}} = 2\, E_{\mathrm{coh}}\, \varphi^{55} = 2\varphi^{50}$ in framework-native units. □

Corollary 2 (Pure φ-power ratios)**.** *All neutrino mass ratios are pure φ-powers of rung differences:*

$$\frac{m_{\nu_i}}{m_{\nu_j}} = \varphi^{\,r_i - r_j}. \tag{3}$$

The sector yardstick, the octave offset, and the SI calibration all cancel in the ratio.

This cancellation is the reason the neutrino sector provides the cleanest falsification targets. The predictions depend on nothing but φ and the rung triple.

4 The quarter-step lattice

Charged fermions sit on integer rungs of the φ-ladder. The neutrino mass splittings are far smaller than the charged-fermion hierarchy, requiring finer resolution. The question is: what is the minimal sub-lattice of $\mathbb{Z}$ that is compatible with the octave structure?

Theorem 3 (Quarter-step resolution)**.** *The minimal refinement of the integer-rung lattice that is compatible with the 8-tick octave period and accommodates a $-1/4$ phase offset is the quarter-step lattice $r \in \frac{1}{4}\mathbb{Z}$.*

Proof. A sub-lattice $r \in \frac{1}{n}\mathbb{Z}$ is *octave-compatible* if n divides $T_{\min} = 8$. The divisors of 8 are $\{1, 2, 4, 8\}$. To accommodate a phase offset of $-1/4$, the lattice must resolve quarter-integer positions: $n \geq 4$. The minimal such divisor is $n = 4$. □

Remark 4 (Frontier status of the phase offset). The $-1/4$ phase offset is the single structural input in the neutrino sector that remains at the frontier. A plausible geometric origin is the C_4 rotational symmetry of each face of Q_3 (each face is a square, providing a natural fourfold subdivision of the integer rung). However, this paper does not claim a closed first-principles derivation of the phase value. The predictions below are conditional on the $-1/4$ offset. If the offset were instead $-1/8$ or $-3/8$, the rung triple and all derived predictions would change. This conditionality is kept visible rather than hidden.

5 Rung spacings from edge partition

The three neutrino masses have two structurally independent rung spacings: $\Delta_{21} := r_2 - r_1$ and $\Delta_{32} := r_3 - r_2$.

Theorem 5 (Rung spacings)**.** *The two spacings are derived from the directional partition of the passive edges of Q_3:*

$$\Delta_{21} = \frac{2^{D-1}}{2} = \frac{4}{2} = 2, \qquad \Delta_{32} = \frac{E_{\text{pass}} - 2^{D-1}}{2} = \frac{7}{2}. \tag{4}$$

The total span is $\Delta_{31} = \Delta_{21} + \Delta_{32} = E_{\text{pass}}/2 = 11/2$.

Geometric origin. Neutrinos, being electrically neutral, couple only to the 1D edge network of Q_3, not to its 2-faces. The total rung span of the neutrino triplet is therefore half the passive edge count: $\Delta_{31} = E_{\text{pass}}/2 = 11/2$. The passive edges partition into two groups:

- **Axial group**: $2^{D-1} = 4$ edges (those along one coordinate axis direction).
- **Remaining group**: $E_{\text{pass}} - 2^{D-1} = 11 - 4 = 7$ edges.

Each group maps to a rung spacing at half-resolution (edge count divided by 2), giving the two spacings. The halving factor reflects the quarter-step lattice: the edge-only sub-lattice operates at $\frac{1}{2}\mathbb{Z}$ resolution, using half the passive edge count.

In quarter-rung numerator coordinates: $\Delta_{21} = 8/4$ and $\Delta_{32} = 14/4$, both elements of $\frac{1}{4}\mathbb{Z}$. The total $\Delta_{31} = 22/4 = 11/2 = E_{\text{pass}}/2$ exhausts the passive edge budget.

6 Absolute baseline rung

The spacings fix the relative positions. The absolute position requires one additional input: the baseline rung r_3 of the heaviest neutrino.

Theorem 6 (Neutrino baseline from hypercube integers)**.** *The integer part of the atmospheric neutrino rung is*

$$r_{\nu_3}^{\text{int}} = -(V + E + F + E_{\text{pass}} + W) = -(8 + 12 + 6 + 11 + 17) = -54. \tag{5}$$

This is the negative sum of all five non-trivial hypercube integers. The active edge count $A = 1$ is excluded because it enters the charged-sector rung formula separately.

The integer baseline -54 is formally verified in Lean 4 (`Masses.BaselineDerivation.neutrino_baseline_eq`).

Theorem 7 (Unique rung triple)**.** *Under the deep-ladder atmospheric confinement (the quarter-rung numerator $4r_3$ lies in the integer interval $(-220, -216)$) and the $-1/4$ phase offset ($4r_3+1 \equiv 0 \pmod 4$), the rung triple is uniquely determined:*

$$(r_1,\, r_2,\, r_3) = \left(-\frac{239}{4},\, -\frac{231}{4},\, -\frac{217}{4}\right). \tag{6}$$

Proof. The integers in $(-220, -216)$ are $\{-219, -218, -217\}$. The phase condition $4r_3 + 1 \equiv 0 \pmod 4$ selects $4r_3 = -217$ uniquely (since $-217+1 = -216$ and $-216 \bmod 4 = 0$, while $-219+1 = -218 \bmod 4 = 2$ and $-218 + 1 = -217 \bmod 4 = 3$, both failing).

Back-propagation through the fixed spacings: $r_2 = r_3 - 7/2 = -217/4 - 14/4 = -231/4$, $r_1 = r_2 - 2 = -231/4 - 8/4 = -239/4$. □

Remark 8 (Phase selection and normal ordering). Among the three deep-window candidates, only the $-1/4$ phase ($4r_3 = -217$) produces normal ordering $m_1 < m_2 < m_3$. The other two candidates produce inverted ordering or degenerate masses inconsistent with oscillation data. The phase $-1/4$ is therefore the unique deep-window quarter-step choice consistent with the observed mass ordering. It remains a frontier item (no deeper cube-geometric derivation is available), but it is selected by data, not freely assumed.

7 The three seam-free predictions

All three predictions below depend only on φ and the rung differences. They are independent of the absolute mass scale, the sector yardstick, the SI calibration anchor τ_0, and any scheme or scale choice in the Standard Model transport layer.

7.1 Prediction 1: Normal mass ordering

Theorem 9 (Normal ordering is forced)**.** *The rung triple $r_1 < r_2 < r_3$ together with $\varphi > 1$ forces*

$$m_1 < m_2 < m_3 \qquad \text{(normal ordering)}. \tag{7}$$

Proof. $m_i = C\,\varphi^{r_i}$ with $C > 0$ and $\varphi > 1$ is strictly increasing in r. Since $-239/4 < -231/4 < -217/4$, the ordering follows. □

Normal ordering is not a choice in this framework. It is a structural consequence. If future experiments decisively establish inverted ordering ($m_3 < m_1$), the rung triple (**??**) is refuted.

7.2 Prediction 2: Squared-mass ratio φ^7

Theorem 10 (φ^7 atmospheric-to-solar ratio)**.**

$$\frac{m_3^2}{m_2^2} = \varphi^{2(r_3-r_2)} = \varphi^{2\times 7/2} = \varphi^7 \approx 29.03. \tag{8}$$

The exponent 7 is the number of remaining passive edges ($E_{\text{pass}} - 2^{D-1} = 11 - 4 = 7$): the same cube integer that controls the atmospheric–solar rung gap.

This ratio is testable once absolute neutrino mass information becomes available from direct kinematic measurements or cosmological surveys.

7.3 Prediction 3: The dimensionless splitting ratio

Theorem 11 (Splitting ratio)**.** *The ratio of mass-squared splittings is*

$$\boxed{R_\Delta := \frac{\Delta m_{31}^2}{\Delta m_{21}^2} = \frac{\varphi^{2\Delta_{31}} - 1}{\varphi^{2\Delta_{21}} - 1} = \frac{\varphi^{11} - 1}{\varphi^4 - 1} \approx 33.82.} \tag{9}$$

Proof. Writing $m_i = C\,\varphi^{r_i}$ for a common pre-factor C:

$$\Delta m_{31}^2 = m_3^2 - m_1^2 = C^2\varphi^{2r_1}(\varphi^{2\Delta_{31}} - 1),$$
$$\Delta m_{21}^2 = m_2^2 - m_1^2 = C^2\varphi^{2r_1}(\varphi^{2\Delta_{21}} - 1).$$

The common factor $C^2\varphi^{2r_1}$ cancels:

$$R_\Delta = \frac{\varphi^{2\times 11/2} - 1}{\varphi^{2\times 2} - 1} = \frac{\varphi^{11} - 1}{\varphi^4 - 1}.$$

Numerically: $\varphi^{11} \approx 199.0$, $\varphi^4 \approx 6.854$, hence $R_\Delta \approx 198.0/5.854 \approx 33.82$. □

The exponents $11 = E_{\text{pass}}$ and $4 = 2^{D-1}$ are the same cube integers that enter the edge subpartition (Theorem **??**). The splitting ratio is therefore a function of the passive edge count and the cube dimension alone.

8 Comparison with oscillation data

Table **??** compares the three seam-free predictions with current experimental constraints.

Table 1: Seam-free predictions vs. current data. These predictions depend only on φ and derived cube integers; they carry no SI-seam dependence.

Prediction	RS value	Experimental	Status
Mass ordering	Normal	Preferred (NuFIT 5.3)	Consistent
m_3^2/m_2^2	$\varphi^7 \approx 29.03$	Not yet measured	Awaiting data
R_Δ	33.82	31.6 ± 1.5 (NuFIT 5.3, NO)	$\approx 1.5\sigma$ tension

The splitting ratio tension is the strongest current pressure on the framework. It is reported transparently here because honest claim hygiene requires it.

Remark 12 (Structural interpretation of the tension). The predicted $R_\Delta = 33.82$ corresponds to $\Delta_{32} = 7/2$. Inverting the splitting ratio formula with the experimental central value $R_\Delta^{\text{exp}} = 31.6$ and holding $\Delta_{21} = 2$ fixed gives $\Delta_{32} \approx 3.36$. The nearest quarter-rung values are $13/4 = 3.25$ and $7/2 = 3.50$. Neither matches 3.36 exactly. The 1.5σ tension is not resolvable by a simple rung shift within the quarter-step lattice: the predicted 7/2 is the unique deep-window quarter-step choice. This makes the tension a genuine quantitative test rather than a knob to be adjusted.

8.1 Absolute mass predictions

Absolute mass predictions require the SI calibration seam (Paper I, Sec. 7). They are therefore less clean than the seam-free predictions above, but they are included for completeness.

Table 2: Absolute neutrino mass predictions (conditional on the SI anchor τ_0 calibrated to $m_e = 0.511$ MeV) vs. NuFIT 5.3 normal-ordering constraints. The $-1/4$ phase offset is a frontier input.

	ν_1	ν_2	ν_3
Rung	$-239/4$	$-231/4$	$-217/4$
m_i^{pred} (eV)	≈ 0.0035	≈ 0.0093	≈ 0.050

Table 3: Neutrino observable predictions vs. NuFIT 5.3 (normal ordering).

Observable	RS prediction	NuFIT 5.3 (NO)
Δm_{21}^2	$\approx 7.3 \times 10^{-5}\,\text{eV}^2$	$(7.53 \pm 0.18) \times 10^{-5}$
$\lvert\Delta m_{31}^2\rvert$	$\approx 2.49 \times 10^{-3}\,\text{eV}^2$	$(2.453 \pm 0.033) \times 10^{-3}$
Σm_ν	$\approx 0.063\,\text{eV}$	$< 0.12\,\text{eV}$ (cosmological)

The predicted sum $\Sigma m_\nu \approx 0.063$ eV is well below the current cosmological upper bound and will be testable by next-generation surveys (CMB-S4, DESI, Euclid). If these surveys establish $\Sigma m_\nu > 0.08$ eV at high confidence, the framework's absolute mass scale would be under pressure.

9 Methods: RS provenance

This section collects framework-specific details for readers interested in the RS provenance.

9.1 Origin of the neutrino baseline

The baseline rung $r_{\nu_3}^{\text{int}} = -54$ is the negative sum of the five non-trivial hypercube integers:

$$V + E + F + E_{\text{pass}} + W = 8 + 12 + 6 + 11 + 17 = 54.$$

The neutrino occupies the deepest ladder position, exhausting the full integer budget of the cube geometry with a minus sign. The active edge count $A = 1$ is excluded because it enters the charged-sector rung formula ($r_e = A + 1 = 2$) separately.

This derivation is formally verified in Lean 4: `Masses.BaselineDerivation.neutrino_baseline_eq`.

9.2 Electroweak sector yardstick

Neutrinos participate in electroweak interactions only, so they use the electroweak sector yardstick $A_{\mathrm{EW}} = 2^{+1} \cdot E_{\mathrm{coh}} \cdot \varphi^{55} = 2\varphi^{50}$ (using $E_{\mathrm{coh}} = \varphi^{-5}$). The sector parameters $(B_{\mathrm{pow}}, r_0) = (+1, 55)$ are from the backbone yardstick table (Paper I, Sec. 5).

9.3 Lean verification

The neutrino sector is formalized in `IndisputableMonolith.Physics.NeutrinoSector` with the following verified objects:

- `res_nu3_simp`, `res_nu2_simp`, `res_nu1_simp`: the three fractional residues.
- `nu3_frac_pred_bounds`, `nu2_frac_pred_bounds`, `nu1_frac_pred_bounds`: mass prediction bounds.
- `dm2_21_frac_pred_in_nufit_1sigma`: Δm^2_{21} within NuFIT 1σ.
- `dm2_31_frac_pred_in_nufit_2sigma`: Δm^2_{31} within NuFIT 2σ.

10 Frontier items

This paper keeps the following items explicitly at the frontier.

Quarter-phase offset. The $-1/4$ phase offset that selects the rung triple from the deep-ladder window is the single non-derived structural input in the neutrino sector. Its boundary status is documented as item B-33 in the framework's derivation roadmap. A plausible geometric origin (the C_4 face symmetry of Q_3) exists but has not been formalized into a closed first-principles argument.

All three seam-free predictions (ordering, φ^7 ratio, splitting ratio) are conditional on this offset. If the offset is wrong, the predictions are wrong. This conditionality is the honest price of publishing a prediction paper before the derivation is complete.

Dirac vs. Majorana character. The framework treats neutrinos as Dirac particles ($Z_\nu = 0$, same φ-ladder as charged fermions). The question of whether neutrinos are Dirac or Majorana is not resolved by the present construction. Neutrinoless double beta decay experiments will address this question independently.

PMNS mixing matrix. A geometric derivation of the PMNS mixing angles from Q_3 path-weight combinatorics is sketched in the framework (`Physics.PMNS.born_mixing`) but is not mature enough for a standalone prediction paper.

11 Falsifiers

Each prediction carries a concrete failure mode.

1. **Mass ordering.** If future experiments (JUNO, DUNE, Hyper-Kamiokande) establish inverted ordering at $> 3\sigma$, the rung triple (**??**) and the entire neutral-sector construction are refuted.

2. **Splitting ratio.** If $R^{\mathrm{exp}}_{\Delta}$ is measured to lie outside the interval (30.0, 37.6) at $> 3\sigma$ (corresponding to $33.82 \pm 3 \times 1.3$, where 1.3 is the framework's structural resolution from adjacent quarter-step alternatives), the edge-partition rung assignment is falsified.

More precisely: if future oscillation data push the central value of R_Δ below ≈ 32.2 at $> 3\sigma$ from 33.82, the framework is refuted. The current 1.5σ tension is the strongest existing pressure.

3. φ^7 **squared-mass ratio.** If absolute neutrino masses are measured (e.g., from KATRIN endpoint analysis combined with oscillation data) and m_3^2/m_2^2 deviates from $\varphi^7 \approx 29.03$ by more than 3σ, the atmospheric–solar rung gap $\Delta_{32} = 7/2$ is refuted.

4. **Sum of masses.** If Σm_ν is measured to exceed 0.08 eV at high significance, the absolute mass scale predicted by the electroweak sector yardstick is under pressure. If $\Sigma m_\nu > 0.12$ eV, the prediction $\Sigma m_\nu \approx 0.063$ eV is strongly disfavored.

5. **Quarter-phase ablation.** If any of the alternative deep-window quarter-step choices ($4r_3 = -219$ or $4r_3 = -218$) produces a splitting ratio closer to experiment, the $-1/4$ phase offset is called into question. Currently, neither alternative produces viable normal-ordering predictions.

12 Discussion

The neutrino sector is the framework's sharpest testing ground for two reasons. First, the neutral-sector simplification ($\mathrm{gap}(0) = 0$) eliminates the charge-band correction entirely, leaving a pure φ-ladder. Second, the splitting ratio R_Δ is dimensionless and depends on nothing but φ and two cube integers ($E_{\text{pass}} = 11$ and $2^{D-1} = 4$). No other sector prediction achieves this level of cleanness.

The current 1.5σ tension between the predicted $R_\Delta \approx 33.82$ and the NuFIT central value 31.6 is the most important open datum for the framework. It is neither comfortable nor fatal. Next-generation oscillation experiments (JUNO, expected to reach sub-percent precision on Δm_{21}^2; DUNE, expected to reach percent-level precision on Δm_{31}^2) will either sharpen the tension into a refutation or relax it into consistency. Either outcome is scientifically productive.

The single frontier item—the $-1/4$ phase offset—is an honest weakness. But it is a weakness that is kept visible and explicitly conditional, not hidden behind suggestive prose. The framework's neutrino predictions are strong enough to be worth publishing even in their conditional form, precisely because they are falsifiable.

13 Conclusion

Given the neutral-sector quarter-step structure of the Recognition Science mass backbone, three predictions follow that are independent of the SI calibration seam:

$$\text{Ordering:} \qquad m_1 < m_2 < m_3 \quad \text{(normal)}, \tag{10}$$

$$\text{Squared-mass ratio:} \qquad m_3^2/m_2^2 = \varphi^7 \approx 29.03, \tag{11}$$

$$\text{Splitting ratio:} \qquad R_\Delta = \frac{\varphi^{11} - 1}{\varphi^4 - 1} \approx 33.82. \tag{12}$$

The splitting ratio is the sharpest near-term test. It uses only φ and the cube integers $E_{\text{pass}} = 11$ and $2^{D-1} = 4$. If future data push R_Δ beyond 3σ from 33.82, the framework is refuted.

The predictions are conditional on one frontier input: the $-1/4$ quarter-phase offset. This conditionality is stated plainly because it is better science to publish a sharp conditional prediction than to wait indefinitely for a complete derivation that may never arrive, or to pretend the condition has already been discharged.

Acknowledgments

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